Sarah Barnes Haddix

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Education

Universität des Saarlandes — Saarbrücken, Saarland, Germany
M.Sc. Computer Science — 10/2026 – Present

The University of North Carolina at Chapel Hill — Chapel Hill, North Carolina, USA
B.S. Computer Science, B.S. Mathematics — 08/2022 – 05/2026

- Graduated with Distinction and Honors; Honors Carolina member
- Major GPA: 3.90/4.00 Overall GPA: 3.79/4.00 Dean's List: 8 semesters

Experience

Owl Cyber Defense — Morrisville, North Carolina, USA
Engineering Intern — 06/2025 – 06/2026

- Developing secure cross-domain network solutions for U.S. government applications using Rust and Java
- Integrating secure chat capabilities into V2CDS, an NCDSMO Baseline Certified technology, using Apache NiFi, streaming XML, and OpenFire
- Developing a secure RDP client/server in Rust for classified environments

The University of North Carolina at Chapel Hill — Chapel Hill, North Carolina, USA
Undergraduate Research Assistant, Real-Time Systems Research Group — 04/2024 – 06/2026

- Conducted research with Dr. James H. Anderson on real-time operating systems, GPU scheduling, and AI; assisted with GPU memory and overload experiments
- Co-authored an ICRA 2025 robotics paper; developed Python/C++ software for a simulated drone using Linux and ROS, and optimized a path-planning neural network using TensorFlow and ONNX

Scale AI — Remote
Software Engineer for AI Training Data — 05/2024 – Present

- Developing programming-focused training data in Java, Python, and Swift to fine-tune large language models

Axai Tech — Cape Town, Western Cape, South Africa
Project Intern — 06/2024

- Developed deep learning models, including LSTM and transformer architectures using PyTorch and Hugging Face, to classify and generate proteins for minimally invasive cancer treatment research

Publications

"On the Necessity of Real-Time Principles in GPU-Driven Autonomous Robots," Syed W. Ali, Angelos Angelopoulos, Denver Massey, Sarah Haddix, et al., *Proceedings of the 2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

Selected Projects & Activities

- **Honors Thesis – Proportional Share Resource Allocation for Real-Time, Time-Shared Systems:** Proved time bounds and correctness properties for the EEVDF scheduling algorithm and analyzed its implementation in the Linux kernel; successfully defended with honors.
- **Wildfire Detection on Autonomous Drones:** Developing a computer vision model for real-time wildfire detection, optimized with TensorRT for Jetson deployment; implementing a messaging protocol for streaming detection data to a remote server.
- **Formal Verification of Onion Franking Using Tamarin:** Formally verified security properties of a metadata-hiding cryptographic protocol for anonymous abuse reporting (COMP 590/637).
- **President, UNC Association for Women in Mathematics:** Organized academic and social events, including study nights and faculty/alumni panels, to foster community in the mathematics department.
- **UNC–Nagoya Cybersecurity Exchange Program:** Selected for a joint undergraduate cybersecurity program exploring cybersecurity, national security, and foreign policy through lectures, workshops, and industry/government visits.